Hans William Alexander Hanley 汉斯·汉隶

AI Safety • Multilingual Representation Learning • Scalable ML Systems

hhanley@cs.stanford.edu hanshanley.com themarginoferror.com
hans-hanley-0694a180 github.com/hanshanley Google Scholar

I research scalable methods for understanding, evaluating, and improving language models, with expertise in multilingual representation learning, semantic modeling, and LLM-based evaluation. My work spans computational social science and AI safety, analyzing information ecosystems, measuring model behavior, and improving AI reliability.

Education

Stanford University **PhD, Computer Science, 2020-2025 | MS, Computer Science, 2020-2024**
GPA: 4.02; Advisor: Zakir Durumeric; Meta Research PhD Fellow; NSF GRFP; Stanford EDGE Doctoral Fellow
University of Oxford **MSc, Statistical Science, 2019-2020 | MSc, Computer Science, 2018-2019**
Distinction (Highest Honors); Daniel M. Sachs Class of 1960 Graduating Scholar at Worcester College
Princeton University **BSE, Electrical Engineering, 2014-2018**
Highest Honors, GPA: 3.982; Information Security & Privacy Concentration; Phi Beta Kappa; Tau Beta Pi; Sigma Xi

Professional Experience

Microsoft AI, Member of Technical Staff **April 2026–Present**
- Built rubric-based LLM-as-a-judge evaluators for Copilot task completion, intent satisfaction, reasoning complexity, response quality, and safety failures. Linked user feedback and behavior to offline evaluations to identify representativeness and failure-mode coverage gaps; presented evaluation methodology and strategy to Microsoft AI's CEO.
- Built production-scale Copilot safety and product-failure signals using LLM labeling, sparse autoencoder (SAE) and embedding-based probes, and distilled classifiers to detect harmful intent (e.g., individual harm), refusal failures, policy violations, and behavioral anomalies; evaluated on production data and used latent-level interpretability to investigate model failures.

Meta, Research Scientist, CAS / Central Applied Sciences **March 2025–April 2026**
- Researched supervised fine-tuning (SFT) techniques and persona-conditioned prompting for LLM digital twins modeling individual behaviors and beliefs; used verbalized sampling to increase response diversity and reduce behavioral homogenization.
- Deployed large-scale multilingual and multimodal LLM pipelines for content understanding, creator classification, and political content and integrity-risk evaluation across Instagram and Threads to inform recommender systems.

Google Research, Health AI Student Researcher **Summer 2024**
- Designed LLM agents for structured querying and high-fidelity reasoning over large clinical datasets, with verification layers for failures under distribution shifts.
- Built evaluations of accuracy, safety, and adversarial robustness for LLM-mediated medical data interactions.

Research Highlights & Selected Publications

Tracking the Takes and Trajectories of News Narratives **USENIX Security 2025**
- Built a narrative-tracking system with LoRA-adapted E5/MPNet encoders and zero-shot stance detection to uncover propaganda and misinformation networks across 4,076 news websites.
- GPU-accelerated DP-Means with FAISS to cluster 428M passages from 29M articles into 146K narratives.

Hierarchical Level-Wise News Article Clustering via Multilingual Matryoshka Embeddings **ACL 2025**
- Developed multilingual Matryoshka embeddings with controllable semantic granularity and level-wise clustering to recover stories, topics, and themes across 54 languages; achieved Pearson $\rho = 0.816$ on extended SemEval 2022 Task 8.
- Made multilingual news clusters interpretable through English summaries: created 38,830 synthetic multilingual multi-document summarization examples and LoRA-fine-tuned Llama 3.1-8B-Instruct, improving ROUGE-1 from 0.332 to 0.414.

TATA: Stance Detection via Topic-Agnostic and Topic-Aware Embeddings **EMNLP 2023**
- Combined topic-aware and topic-agnostic embeddings with triplet/contrastive learning, retrieval-mined pairs, and synthetic data for unseen-topic stance detection; achieved 0.771 zero-shot macro-F1 on VAST.

Across the Firewall: Foreign Media's Influence on Chinese Social Media **PNAS 2025**
- Built multilingual MPNet/FAISS retrieval and trained three bilingual mDeBERTa paraphrase models to link Weibo narratives to 8.26M articles; used NETINF to quantify Russian media's outsized influence.

Skills

Programming	Python, Java, C, C++, C#, Go, SQL, R, MATLAB; Git; PyTorch, TensorFlow, Hugging Face, FAISS
ML/LLM	LoRA/PEFT; contrastive learning; multilingual NLP; synthetic data; LLM eval; feature attribution; safety/robustness
Languages	English (native); Mandarin Chinese (HSK 2.0, Level 5)

Selected Awards & Honors

- CSCW 2025 Best Paper Honorable Mention (Top 4%).
- Rising Stars in Data Science (Stanford/UCSD/UChicago); Meta Research PhD Fellow; NSF Graduate Research Fellowship (GRFP); Stanford EDGE Doctoral Fellow; University of Oxford Daniel M. Sachs Class of 1960 Graduating Scholar.
- Charles Ira Young Memorial Tablet & Medal; SEAS Hayes–Palmer Prize; George B. Wood Legacy Prize.